Official responses to questions supersede original competition statements as well as any earlier question responses where there is contradiction. The questions are numbered sequentially as responded to, where Answer 1 is the earliest response.

Information about the ASME EFx events can be found at: https://efests.asme.org/

PODIUM Q&A Update October 13, 2026

Question 1:

- When designing the end-effector, can it be modular in terms of having multiple swappable designs for each event or is the intent simply having one design that fits all?

Answer 1:

- Yes. Swappable end effectors are allowed, within these limits:
    - End effectors are part of the Robot, whether or not they are attached, and they are not spare parts.
    - An end effector is only the tool at the end of a mechanism, such as a gripper or a hook. Other parts of the Robot may not be swapped (repairs under Rule 8-3 are still allowed).
    - An end effector must attach and detach without tools (for example with a latch, pin, or quick-release) through a connection already on the Robot. No other change may be made to the Robot to fit one.
    - All end effectors are inspected with the Robot at its inspection, which happens at the team's first Discipline (Rule 8-1). Only inspected end effectors may be used, and Rule 8-3 applies to any change after inspection.
    - With any end effector fitted, the Robot must meet every Robot rule, including fitting within the 40 cm cube at the start of each Run, weighing 15 kg or less, and staying within the 75 cm and 120 cm expansion limits (Section 7.1).
    - Because the Robot must fit within the 40 cm cube at inspection (Rule 7.1-2), the Robot and all of its end effectors must fit together inside the team's 40 cm container.
    - With each end effector fitted, the Robot must also meet every Robot rule: it must fit within the 40 cm cube at the start of every Run (Rule 7.1-2), weigh 15 kg or less as it competes (Rule 7.1-4), and stay within the 75 cm and 120 cm expansion

limits (Rule 7.1-3). The Robot is weighed with each end effector fitted, one at a time; end effectors that are not fitted do not count toward the 15 kg limit.

- Swaps may be made between Runs at the team table (Rule 8-3) or during the 60-second setup at the Station (Rule 6.3-1), but never during a Run or while moving between Stations. Teams will often go directly from one Discipline to the next: the Robot goes into its container as soon as a Run ends and comes out only for the next setup. In that case, any swap must happen within the 60-second setup, and no extra time is given (Rule 6.3-1).

Question 2:

- What are some more details on the dimensions of the climbing wall, such as the spacing of the rungs/holds?

Answer 2:

- The Ascent climbing structure is the same at every event:
  - It is an open ladder of uniform horizontal rungs, with no other holds or bars and no wall face behind the rungs. The bracing that holds the ladder upright is out of the Robot's reach.
  - Rungs are round, 25 mm in diameter (20 to 30 mm permitted, the same for every rung on a ladder).
  - Rungs are spaced 20 cm on center. The first rung is 20 cm above the floor, and the top rung is at 2.2 m.
  - The climbable width is at least 120 cm.
  - The ladder stands on the far edge of the 3.6 × 3.6 m Field, with padding at least 40 mm thick covering the floor at least 60 cm in front of it.
  - Height Bands are marked at 0.5, 1.0, and 1.5 m, and the Summit is at 2.0 m, measured from the floor to the lowest point of the Robot (Section 12.5).
  - Materials vary by event, since each host builds with locally available materials. The dimensions above are the same everywhere.
  - No drawing will be published. The dimensions above are the official specification.

Question 3:

- Will we be given the dimensions of the starting zone?

Answer 3:

- Yes. In every Discipline the Starting Zone is a marked 60 × 60 cm square. Your Robot must start inside it, within its 40 cm starting size (Rules 7.1-2 and 11.3-2). Its location in each Field:
  - The Sprint: at the start line, centered on the lane's width.
  - The Lift: behind the loading zone, centered on the Field's width.

- The Cycle: behind the loading area, centered on the Field's width (as shown in Figure 4).
- The Marksman: behind the No-Cross Zone line, centered on the Field's width.
- The Ascent: on the floor in front of the climbing structure (Section 12.5).

Question 4:

- Will the hurdles be bolted down?

Answer 4:

- Not necessarily. Each host secures the Hurdles and Gates using its own method and material, but their size and position are the same everywhere:
  - Each Hurdle is a bar 60 cm long, 5.0 cm tall, and 2 to 4 cm thick in the direction of travel, with its end 20 cm from the lane edge.
  - Each Gate is two posts, 5 × 5 cm and 30 cm tall, with a 60 cm clear opening between them and a 20 cm gap between the lane edge and the near post.
  - Robots must avoid bumping into Hurdles and Gates. They are not permanent fixtures, so contact can push them out of place. Hurdles are to be crossed by driving over them (Section 12.1), and Gates passed through, without moving either. Once a Discipline's first Run starts, the Field is accepted as it is, and Runs are not replayed for Field conditions, including a Hurdle or Gate that shifted (Rule 6.4-1).
- Moving a Hurdle or Gate out of place is interference with the Field and is penalized under Section 10.1. An accidental move draws a Warning; repeated or deliberate moves draw escalating penalties, and significant interference voids the Run. The Head Judge decides what counts as significant. Damaging the Field is not allowed (Rules 7.4-2 and 10-2).

Question 5:

- Is there a wall behind the target?
- Is there a wall below the tall goal as showed in the image?

Answer 5:

- Both goals stand on the Field's far edge (Section 12.3). Each goal catches the Balls that pass through its opening and returns them to the loading area. The openings are the same at every event:
  - Low Goal: 40 × 40 cm at floor level, its center 2.50 m from the Field's left edge (as seen from the Starting Zone).

    - High Goal: a 30 cm ring centered at 1.0 m high, its center 0.90 m from the Field's left edge.

- Behind the goals, a backstop (a net or a panel) runs along the Field's far edge, so a Ball that misses a goal and hits the backstop stays in play. The structure around and below the High Goal is built by each host and may be a panel or a frame. No wall or net is required along the Field's other edges. A Ball may miss a goal, but it may not leave the Field: penalties under Section 10.1 apply to a Ball that leaves the Field.

Question 6:

- Can a singular ball can score multiple points in the cycle?

Answer 6:

- Yes. A Ball scores every time it passes fully through a goal opening: 2 points through the Low Goal and 5 points through the High Goal, doubled in the Autonomous Phase (Section 12.3). Scored Balls roll back into the loading area through an open return trough that runs inside the Field along its left edge, so the same Ball can be collected and scored again.

- Robots may collect Balls anywhere on the Field floor, including Balls that missed a goal, but not from the return trough. A Robot may control at most three Balls at a time (Section 12.3), a Ball that leaves the Field draws a penalty (Section 10.1), and the Discipline Score is capped at 140 points.

Question 7:

- Will we get the dimensions of the scoring rings within the target?

Answer 7:

- Yes. The Target is 60 cm across, with five concentric rings of equal 6 cm width: outer radii of 6, 12, 18, 24, and 30 cm, scoring 20, 16, 12, 8, and 4 points from the bullseye outward. It lies flat on the floor on a thin mat (5 mm or less), its center 2.50 m from the far edge of the No-Cross Zone line and centered on the Field's width. A Bean Bag scores by the ring it rests in, with more than half of it inside (Sections 3.3 and 12.4).

Question 8:

- Are bean bags cleared between tosses?

Answer 8:

- No. Bean Bags are not cleared or returned during a Run: each one stays where it comes to rest until the Run ends. The Field is reset between Runs, with all eight Bean Bags returned to the loading station (Section 12.4).

Question 9:

- Can you measure your robot's position relative to the target before starting?

Answer 9:

- No. Teams may not approach or touch a Field (Rule 6.4-1), so they may not measure it. The only exception is the Crew placing the Robot in the Starting Zone during setup (Rule 11.3-2). If you are in doubt about any dimension, ask an Official to check it. Concerns about a Field must be raised before that Discipline's first Run begins; after that, the Field is accepted as it is (Rule 6.4-1).

- The Target's position is fixed and the same at every event: its center is 2.50 m from the far edge of the No-Cross Zone line and centered on the Field's width, and the Starting Zone is a marked 60 × 60 cm square behind the No-Cross Zone line, centered on the Field's width.

- During the 60-second setup (Rule 6.3-1), the Crew may place and align the Robot anywhere within the Starting Zone, but nothing other than the Robot may be placed on or extended into the Field: no tapes, jigs, or markers (Rules 6.4-1 and 11.3-2). The Robot's own onboard sensors may be used freely.

Question 10:

- The rules say there are rungs, so are there continuous ladder or is it a mix of holds?

Answer 10:

- See Answer 2.

Question 11:

- What is the spacing between rungs and also is the spacing open or closed off?

Answer 11:

- See Answer 2.

Question 12:

- Can we use a grappler? (Does that count as an extension of the robot)

Answer 12:

- Yes. A Robot may grip and attach to the Ascent climbing structure (Section 3.3, Structure; Rule 10-4). A grappler is part of the Robot: it must stay attached to the Robot at all times and may never be a separate robot or device (Rule 7.1-1). The Robot, including the grappler and any line attached to it, must stay within the expansion limits at

all times: 75 cm in any horizontal direction and 120 cm in height (Rule 7.1-3). Any launched part must stay low, safe energy (Rule 7.4-2), and a Robot may never grapple another Robot (Rule 10-4).

Question 13:

- What will the dimensions be for the wall in the Ascent challenge (width & Length)? As well as the general geometry/size for the rungs, holds, and bars that are distributed throughout the wall?

Answer 13:

- See Answer 2.

Question 14:

- I wanted to ask about whether there are going to be multiple robots competing in a run at the same time, and on the same field. I am asking this because section 10.3 says that a robot cannot interfere with another robot's scoring, but I cannot find any mention in the rules/description of the disciplines that there will be multiple robots performing a run at the same time.

Answer 14:

- No. In qualification, each Run is one Robot alone on its Field (Section 3.1, Run). In the playoff, each partner Robot also takes its own Run (Rule 14-5), and matches may run head to head on two identical Fields side by side, or one Pair at a time on a single Field (Rule 14-6). Two Robots never share one Field. Rule 10-3 still applies wherever a Robot or its control link could affect another team, for example on side-by-side Fields or adjacent Stations: no defense, interference, or contact of any kind is permitted.

Question 15:

- I have some questions regarding the dimensions and materials of the elements for the Sprint that are not present in the rules. What is the height of the gate posts? How thick are the hurdles? (Not the 60 cm width or the 5cm height, the thickness of hurdles in the direction of 7 m side of the lane). What material are the hurdles made out of?

Answer 15:

- See Answer 4.

Question 16:

- I would like to seek clarification regarding Rule 8.3, specifically the distinction between “minor repairs and adjustments” and “major modifications.”

Are teams permitted to make changes to the machine between different rounds/disciplines, provided that the changes do not fundamentally alter the machine's design or functionality?

To better understand the intended interpretation of Rule 8.3, could you please classify the following examples as either “minor adjustment/allowed” or “major modification/not allowed”?

1. A specific link/component is originally positioned horizontally, but its orientation is slightly changed for a particular discipline.

2. A shaft inside a cylindrical assembly is moved slightly inward for two different disciplines, with no other change to the assembly.

3. The overall chassis dimensions are increased between two different disciplines.

4. A mounting point or bracket is repositioned by a small amount to accommodate a different configuration required for a particular discipline, without changing the primary mechanism.

5. A component is replaced with another component having the same function and similar specifications, but with a slightly different geometry or mounting arrangement.

If these examples are not sufficient, it would be helpful if you could provide a few examples of what would specifically constitute a “minor repair/adjustment” versus a “major modification” under Rule 8.3.

This clarification would help teams understand what changes can be made between disciplines while remaining compliant with Rule 8.3.

Thank you.

Answer 16:

- Between Runs, Rule 8-3 allows repairs, replacing broken parts with equivalent parts, adjustments that keep the Robot as it was (for example tightening, re-aligning, or recalibrating), and code changes. Any change to the Robot's fit, form, or function is a major modification and is not allowed, and setting the Robot up differently for a particular Discipline is a change to its function. Swapping between inspected end effectors through a tool-free connection is not a modification (see Answer 1). Your examples:
    - Example 1: Not allowed.
    - Example 2: Not allowed.
    - Example 3: Not allowed.
    - Example 4: Not allowed.
    - Example 5: Not allowed. A different geometry or mounting changes the Robot's form or fit. Replacing a broken part with an equivalent part is still allowed.
- Any change that could affect the Robot's size, weight, or safety requires re-inspection before its next Run (Rule 8-3).

Question 17:

- I would also like to seek clarification regarding Rule 12.5 the elements installed on the climbing wall.

  The rulebook states that the wall will be fitted with “rungs, holds, and bars.” Could you please provide more information about these elements?

  Specifically:

  1. What exactly is meant by “rungs,” “holds,” and “bars” in this context?

  2. What material(s) will these elements be made from?

  3. What are their approximate width and cross-sectional dimensions?

  4. How far do they project from the wall (depth)?

  5. What is the vertical spacing/height difference between consecutive rungs/bars/holds?

  6. Will the wall have open/hollow sections, similar to a conventional ladder, or will the climbing elements be attached to a solid/rigid wall structure?

  7. At the summit, will there be a flat top/platform onto which the robot can climb and rest, or will the robot be required to reach and hold onto the topmost rung/bar?

  If these specifications are to be determined by the host university, it would still be helpful to have some base/reference dimensions, drawings, or example images that the hosts are expected to follow.

  These details have a significant impact on the mechanical design of the robot. For example, whether the climbing structure is an open ladder-like frame vs. a solid wall with protruding elements changes the required hooking, gripping, engagement, and climbing mechanisms considerably.

  An isometric/3D reference image or model of the climbing wall and its elements would therefore be particularly helpful.

  Thank you.

Answer 17:

- Parts 1 to 6: see Answer 2.

- Part 7: There is no platform at the top. The ladder's top rung is at 2.2 m, and the Summit is reached when the Robot holds on the climbing structure with its lowest point at 2.0 m or higher and nothing touching the floor (Section 12.5). The safety-line mast above the ladder is not part of the climbing structure and may not be used for climbing.

- As noted in Answer 2, no drawing will be published.

Question 18:

- I have some questions about the dimensions of elements in Ascent that I couldn't find in the rules. What is the width of the ladder? What is the diameter of the rungs? What is the height increment between each rung?

Answer 18:

- See Answer 2.

Question 19:

- What is the weight and dimensions of the dumbbells?

Answer 19:

- There are three masses: Light 1 kg, Standard 2 kg, and Heavy 4 kg (Section 11.2), each within 5% of its marked mass. Each is a coated weight (vinyl, rubber, or similar) with a central hand grip, about 15 to 20 cm across at its largest dimension, and color coded by mass. Hosts may use any locally available dumbbell within these limits, so the exact shape and color may vary by event. For each Run, six of each mass stand upright in the loading zone, grouped Light, Standard, Heavy from left to right as seen from the Starting Zone.

Question 20:

- We are designing a climbing mechanism for the Ascent event. The rules state that the 2.2 m wall is fitted with rungs, holds, and bars, but they do not specify their dimensions, spacing, or arrangement. Are these features placed in a standardized layout, and will an official field diagram with exact locations be provided?

Answer 20:

- See Answer 2.

Question 21:

- Will the Ascent's structure be made of steel, aluminum or some other material?

Answer 21:

- See Answer 2.